

Layer 4 — Knowledge, reasoning, and writing automation

Product-owner version

Product goal: turn the research platform's outputs into usable knowledge, reasoning context, and writing material.

Layer 1 prepared the runtime.

Layer 2 prepared the roles.

Layer 3 prepared scientific execution loops.

Layer 4 prepares the **thinking layer** that sits above experiments but below full orchestration.

It answers:

Can I query my research notes, experiment outputs, papers, code summaries, and mechanism reports?

Can I turn runs into structured reasoning context without manually reading every artifact?

Can OpenClaw or a local/remote model help me reason over the PKM without corrupting the vault?

Can outputs from GRN/NCA/ART runs become Zettelkasten notes, paper sections, or next-experiment ideas?

Layer 4 boundary

Layer 4 should do:

prepare PKM context indexes

prepare OpenClaw workspace

prepare safe local/remote model reasoning configs

prepare query smoke tests

Layer 4 should not do:

- not replace the Zettelkasten vault
- not overwrite notes
- not auto-generate final paper sections
- not run GRN simulations
- not train NCA models
- not launch Runpod jobs
- not build Agentfield controller
- not build Paperclip adapter

The rule is:

- Bundle 9 owns the PKM structure.
- Layer 4 lets tools reason over that PKM safely.

So this layer does not reorganize the vault again. It adds **reasoning access**.

Bundle inside Layer 4

Bundle 8 – OpenClaw + PKM reasoning workspace

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Product outcome

A prepared workspace where OpenClaw, local model clients, remote model clients, and lightweight indexing tools can reason over selected research material.

This should connect:

Atomic Zettelkasten

source notes, atoms, molecules, research questions, alloy notes

NCA-ART-GRN artifacts

mechanism reports, search reports, candidate DSL files, figures, metrics

Remote model brain endpoint

thin model client from Layer 1

OpenClaw workspace

tool reasoning, query workflows, local context packs

Concretizations

prepare_openclaw_pkm_workspace

check_openclaw_workspace

prepare_pkm_context_index

prepare_local_model_reasoner_config

prepare_pkm_query_smoke_test

Potentially later:

prepare_experiment_report_ingest

prepare_zettelkasten_reasoning_bridge

prepare_next_experiment_question_generator

prepare_mechanism_report_to_alloy_note_bridge

Product meaning of each concretization

`prepare_openclaw_pkm_workspace`

Creates the workspace where OpenClaw-style reasoning workflows live.

Suggested path:

```
/workspace/repos/openclaw-workspace/
```

It should contain:

```
configs/  
contexts/  
queries/  
tools/  
runs/  
smoke_tests/
```

Product value: gives AI reasoning tools a home separate from the research repo and separate from the PKM vault.

`check_openclaw_workspace`

Checks that the OpenClaw workspace exists and is usable.

It should report:

```
workspace path  
config path  
context path  
query path  
runs path
```

model config present or missing

Product value: lets the operator verify the reasoning workspace before model calls or indexing.

prepare_pkm_context_index

Creates an index or manifest over selected PKM and artifact material.

It should index or list selected paths such as:

```
/workspace/pkm/zettelkasten/20_knowledge_kasten  
/workspace/pkm/zettelkasten/30_publish_kasten  
/workspace/pkm/zettelkasten/40_experiments  
/workspace/artifacts/nca-art-grn/mechanism_reports  
/workspace/artifacts/nca-art-grn/search_reports
```

It should not index everything by default.

Product value: lets tools ask useful questions over the research corpus without dumping the whole vault into a prompt.

prepare_local_model_reasoner_config

Creates model-routing config for local or remote reasoning.

It should connect to Layer 1's remote model client contract:

```
local code -> remote model -> response
```

But it should support multiple reasoning profiles:

```
fast_summary  
deep_research_reasoning  
paper_outline
```

```
mechanism_review
next_experiment_suggestion
codebase_triage
```

Product value: makes the model reasoning layer reusable by OpenClaw, later Agentfield agents, and manual CLI workflows.

prepare_pkm_query_smoke_test

Creates a tiny query test that asks against a small selected context pack.

Example smoke questions:

```
What mechanism reports exist?
Which candidate has the strongest perturbation evidence?
Which Zettelkasten research questions are linked to NCA-ART?
Which mechanism reports should become alloy notes?
What is the next experiment suggested by the latest search report?
```

Product value: verifies that PKM reasoning works without modifying notes or launching experiments.

Layer 4 relationship to earlier layers

```
Layer 1
  provides runtime and remote model client

Layer 2
  provides PKM vault and Publisher/AI Engineer roles

Layer 3
```

produces mechanism reports, search reports, candidate artifacts

Layer 4

reasons over those notes and artifacts

Layer 4 relationship to later layers

Layer 4 prepares reasoning tools that later become useful to Agentfield and Paperclip:

Agentfield later

can call reasoners to summarize experiments, suggest next runs,
rank hypotheses, and triage failures

Paperclip later

can show model-generated summaries, review tasks,
next-experiment proposals, and paper-writing suggestions

But at this stage:

OpenClaw/PKM reasoning is manual or config-smoke driven.

Agentfield orchestration comes later.

Paperclip dashboard comes later.

Layer 4 success condition

After Layer 4, you should be able to say:

My PKM vault is structured.

My experiment reports are stored.

My OpenClaw workspace exists.
My reasoning config points to local/remote model clients.
I can run a safe query smoke test over selected research context.
No notes are overwritten.
No experiments are launched.
No papers are auto-generated.

In one line:

Layer 4 turns research outputs and PKM notes into queryable reasoning context.